

1. True/False: Suppose you learn a word embedding for a vocabulary of 60000 words. Then the embedding vectors could be 60000 dimensional, so as to capture the full range of variation and meaning in those words.

1 / 1 point

☐ True

☒ False

✓ **Correct**

No, the dimension of word vectors is usually smaller than the size of the vocabulary. Most common sizes for word vectors range between 50 and 1000.

2. What is t-SNE?

1 / 1 point

☐ A linear transformation that allows us to solve analogies on word vectors

☒ A non-linear dimensionality reduction technique

☐ A supervised learning algorithm for learning word embeddings

☐ An open-source sequence modeling library

✓ **Correct**

Yes

3. Suppose you download a pre-trained word embedding which has been trained on a huge corpus of text. You then use this word embedding to train an RNN for a language task of recognizing if someone is happy from a short snippet of text, using a small training set.

x (input text)	y (happy?)
Having a great time!	1
I'm sad it's raining.	0
I'm feeling awesome!	1

Even if the word “wonderful” does not appear in your small training set, what label might be reasonably expected for the input text “I feel wonderful!”?

☒ y=1

☐ y=0

☒ **Correct**

Yes, word vectors empower your model with an incredible ability to generalize. The vector for “wonderful” would contain a negative/unhappy connotation which will probably make your model classify the sentence as a “1”.

4. Which of these equations do you think should hold for a good word embedding? (Check all that apply)

1 point

☐ $e_{man} - e_{woman} \approx e_{aunt} - e_{uncle}$

☒ $e_{man} - e_{woman} \approx e_{uncle} - e_{aunt}$

✔ Correct

The order of words is correct in this analogy.

☐ $e_{man} - e_{uncle} \approx e_{woman} - e_{aunt}$

☐ $e_{man} - e_{aunt} \approx e_{woman} - e_{uncle}$

You didn't select all the correct answers

5. True/False: The most computationally efficient formula for Python to get the embedding of word 1021, if C is an embedding matrix, and o_{1021} is a one-hot vector corresponding to word 1021, is $C^T * o_{1021}$.

1 / 1 point

☐ True

☒ False

✔ Correct

It is computationally wasteful because the element-wise multiplication will be extremely inefficient.

6. When learning word embeddings, we create an artificial task of estimating $P(\text{target} \mid \text{context})$. It is okay if we do poorly on this artificial prediction task; the more important by-product of this task is that we learn a useful set of word embeddings.

1 / 1 point

☒ True

☐ False

✓ Correct

7. In the word2vec algorithm, you estimate $P(t \mid c)$, where t is the target word and c is a context word. How are t and c chosen from the training set? Pick the best answer.

1 / 1 point

☐ c is the one word that comes immediately before t .

☐ c is a sequence of several words immediately before t .

☐ c is the sequence of all the words in the sentence before t .

☒ c and t are chosen to be nearby words.

✓ Correct

8. Suppose you have a 10000 word vocabulary, and are learning 100-dimensional word embeddings. The word2vec model uses the following softmax function:

1 / 1 point

$$P(t \mid c) = \frac{e^{\theta_t^T e_c}}{\sum_{t'=1}^{10000} e^{\theta_{t'}^T e_c}}$$

True/False: After training, we should expect θ_t to be very close to e_c when t and c are the same word.

- ☐ True
- ☒ False

✓ **Correct**

To review this concept watch the *Word2Vec* lecture.

9. Suppose you have a 10000 word vocabulary, and are learning 500-dimensional word embeddings. The GloVe model minimizes this objective:

1 / 1 point

$$\min \sum_{i=1}^{10,000} \sum_{j=1}^{10,000} f(X_{ij})(\theta_i^T e_j + b_i + b_j' - \log X_{ij})^2$$

True/False: X_{ij} is the number of times word j appears in the context of word i .

- ☒ True
- ☐ False

✓ **Correct**

X_{ij} is the number of times word j appears in the context of word i .

1 / 1 point

10. You have trained word embeddings using a text dataset of m_1 words. You are considering using these word embeddings for a language task, for which you have a separate labeled dataset of m_2 words. Keeping in mind that using word embeddings is a form of transfer learning, under which of these circumstances would you expect the word embeddings to be helpful?

☒ $m_1 \gg m_2$

☐ $m_1 \ll m_2$

✓ Correct